

HICHAM TBINA

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📍 Morocco-Rabat

Electrical Engineering and Industrial Systems Control

PROFILE

Currently in my 3rd year of engineering studies in Electrical Engineering and Industrial Systems, seeking a final-year internship (PFE) in automation, power electronics, and embedded systems. With strong skills in electrical design, industrial automation, and system integration, I am eager to contribute to innovative projects and apply my expertise in real-world applications. I welcome the opportunity to join your team.

SKILLS

- Flexibility at Work
- Team work
- Project management
- Technical Communication

TECHNICAL SKILLS

- Matlab, simulink
- C++ and python coding
- VHDL / FPGA
- PID, MPC, adaptive and optimal control
- AI and ML/DL
- Spice (LTspice/PSpice)
- Power electronics
- System modeling
- Signal processing

EDUCATION

Higher Normal School of Technical Education (ENSET): 2023-2026 Mohammedia
Electrical Engineering and Industrial Systems Control

Preparatory Classes for Higher Technician Certificate (BTS): 2021-2023 sala

Electrotechnics

Baccalaureate Diploma : 2021 rabat

Electrical Science

PROFESSIONAL EXPERIENCE

club electro lab : from 1/01/2024 to present

A university club participating in robotics competitions and developing electrical and control systems with innovative.

Learn R&D : from 07/07/2025 to 04/09/2025 rabat

A 2-month engineering internship at LEAR, focused on studying an ideal diode circuit and performing Worst Case Analysis (WCA), including VBA automation of the WCA simulation, to optimize the reliability and performance of embedded electronic systems.

Redal-BCC : from 24/07/2025 to 24/09/2025 rabat

A 2-month introductory internship at REDAL, focused on the SCADA system migration project from Power CC to LYNX, aimed at modernizing the supervision and management of the electrical network.

post source AGDAL : from 07/07/2022 to 07/08/2022 rabat

My 2-month internship at REDAL provided valuable experience in electrical engineering. Working alongside industry professionals, I gained practical skills in troubleshooting and problem-solving.

Projects

- **Smart Microgrid with PLCnext** : Real-time hybrid microgrid supervision (IMEON 3.6). Programmed PLCnext AXC F2152 (Ladder, Modbus TCP/IP, HMI, Node-RED).
Skills: PLC, Modbus, HMI, data visualization.
- **UPS Design** : Switched-mode supply with lithium BMS and PWM inverter.
Skills: Power electronics, inverter control, filtering.
- **Collaborative Robotic Arm (Cobot)** : Safe human-robot operation using force & proximity sensors.
Skills: Control, safety systems, sensor integration.
- **Variable Voltage Regulator** : PID controller for a DC/DC converter for a stable output voltage.
Skills: converter design, transient control, optimization.
- **6-DoF Reusable Launch rocket Simulation** : Full flight modeling (PID, MPC) with MATLAB/Simulink.
Skills: Flight dynamics, advanced control, sensor/actuator modeling.
- **PV System with Nonlinear Backstepping Control** : PV + bidirectional converter with MPPT and CC/CV battery management.
Skills: Nonlinear control, Zeta converter, energy coordination.